Lab 1: Requirements Engineering

Requirements Table - Airport Lost Luggage Claim & Tracking Portal

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Problem Statement #30: Smart Cities, Transport & Logistics

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale	Comments
FR-001	Functional	The system shall cross-reference passenger lost baggage reports against unrouted luggage scan logs across all connected airport terminals.	High	Pass if a matching bag tag generates a recovery alert; fail if an invalid bag tag is accepted.	Helps locate missing baggage automatically.	Check that the bag tag format and matching condition are clear.
FR-002	Functional	The system shall allow a passenger to register a lost baggage claim using the baggage tag number, flight number, travel date, and contact details.	High	Pass if a valid claim is stored and assigned a unique claim ID; fail if the claim is saved without mandatory details.	Provides an official record of the lost baggage case.	Mandatory fields and the unique claim ID make this testable.
FR-003	Functional	The system shall display the complete scan timeline of a baggage item across all connected airport terminals.	High	Pass if the system displays the airport, timestamp, and scan status for every available scan; fail if an available scan is omitted.	Allows passengers and agents to track baggage movement.	The acceptance criterion specifies the information that must be shown.
FR-004	Functional	The system shall allow a baggage service agent to update the status of a baggage claim.	High	Pass if an authorized agent can change the status to Received, Located, Delivered, or Closed; fail if an unauthorized user changes it.	Keeps claim information accurate and up to date.	Authorization should be checked during testing.
FR-005	Functional	The system shall initiate a compensation request when baggage is confirmed as lost according to airline policy.	Medium	Pass if an eligible claim creates a compensation request with the correct claim ID and passenger details; fail if an ineligible claim creates one.	Automates the compensation process for passengers.	The airline eligibility policy should be available to the system.
NFR-001	Nonfunctional - Performance	The baggage tracking timeline API shall return the complete multi-leg scan history within 200 milliseconds under normal operating conditions.	High	Pass if at least 95% of 1,000 test requests return within 200 milliseconds; fail otherwise.	Ensures that baggage tracking feels fast and responsive.	The response-time target is measurable.
NFR-002	Nonfunctional - Security	The system shall restrict baggage claim and compensation information to authenticated and authorized users.	High	Pass if unauthenticated users are denied access and agents can access only permitted functions; fail if restricted information is exposed.	Protects passenger personal and financial information.	Test both unauthenticated access and agent permissions.